

A disproof of Conjecture 00000002149

The Colin de Verdière bound $\mu(\overline{G}) \leq |V| - \omega(G) - 1$ fails

1 The conjecture

Definition: The Colin de Verdière parameter $\mu(G)$ is the spectral-rigidity invariant of a graph. *Conjecture:* $\mu(\overline{G}) \leq |V| - \omega(G) - 1$ (an upper bound for the complement); the bound is tight (attained by the complete graph with empty complement).

Reading. For every finite simple graph $G = (V, E)$, $\mu(\overline{G}) \leq |V| - \omega(G) - 1$, where ω is the clique number; moreover equality holds for complete graphs. We refute the inequality (and with it the tightness clause) on two infinite families: the complete graphs K_n , $n \geq 1$ (the case the conjecture itself names), and the edgeless graphs $\overline{K}_n$, $n \geq 2$. Among complete graphs, the only one on which the claimed tightness holds is the order-zero graph K_0 (with the convention $\mu(K_0) = -1$), which is excluded here ($n \geq 1$).

2 Definitions (as in the Lean file)

Definition 1 (Colin de Verdière parameter). For a graph G on a finite vertex set V , a real matrix $M \in \mathbb{R}^{V \times V}$ is *admissible* (`IsCdVMatrix`) if

- M is symmetric;
- (M1) for $i \neq j$: $M_{ij} < 0$ if $ij \in E$, and $M_{ij} = 0$ if $ij \notin E$ (no condition on the diagonal);
- (M2) M has exactly one negative eigenvalue, counted with multiplicity: the number of indices i with $\lambda_i < 0$ is 1, where (λ_i) is Mathlib’s eigenvalue family `IsHermitian.eigenvalues` (listed with multiplicity);
- (M3) Strong Arnold Hypothesis: every symmetric X with $X_{ii} = 0$ for all i , $X_{ij} = 0$ for $ij \in E$, and $MX = 0$ is the zero matrix.

The corank is $|V| - \text{rank } M$ (`corank`), and $\mu(G)$ is the largest corank of an admissible matrix (`cdvMu`, a supremum in $\mathbb{N}$; the set of coranks is bounded by $|V|$, so every admissible M gives $\text{corank } M \leq \mu(G)$, lemma `le_cdvMu`). In (M3), the source’s condition “ $X_{ij} = 0$ if $i = j$ or $M_{ij} \neq 0$ ” is the same as ours, because by (M1) $M_{ij} \neq 0$ for $i \neq j$ exactly when $ij \in E$. The clique number is Mathlib’s `SimpleGraph.cliqueNum`.

Convention. μ is $\mathbb{N}$ -valued in Lean, so the inequality $\mu(\overline{K}_n) \geq 0 > -1$ alone would hold for type reasons. For that reason the main theorem also exhibits an explicit admissible matrix in each case, and it states a nontrivial lower bound ($\mu(\overline{K}_n) \geq 1$ for $n \geq 2$; $\mu(K_n) \geq n - 1$ for $n \geq 2$).

3 Proof

Lemma 2 (`card_neg_eig_diagonal`). *The number of negative eigenvalues of a diagonal matrix $\text{diag}(d)$ equals the number of negative entries d_i .*

Proof. The roots of the characteristic polynomial $\prod_i (X - d_i)$ are the multiset $\{d_i\}$, and they are the eigenvalues listed with multiplicity (Mathlib’s `roots_charpoly_eq_eigenvalues`). $\square$

Theorem 3 (Family (a): $G = K_n$). *Let $n \geq 1$. Then $\omega(K_n) = n$ (`cliqueNum_top`), so the claimed bound is -1 . The matrix $M = \text{diag}(-1, 0, 1, \dots, 1)$ is admissible for the edgeless graph $\overline{K_n}$ (`isCdV_empty`), and for $n \geq 2$ its corank is 1 (`corank_empty`). Hence $\mu(\overline{K_n}) \geq 1 > -1$ for $n \geq 2$; for $n = 1$, $M = (-1)$ is admissible, so the defining set is nonempty and $\mu(\overline{K_1}) \geq 0 > -1$.*

Proof. (M1): off-diagonal entries are 0 and there are no edges. (M2): exactly one entry is negative. (M3): $(MX)_{ij} = d_i X_{ij}$, so $X_{ij} = 0$ whenever $d_i \neq 0$, i.e. whenever $i \neq 1$ (index 1 carries the 0). For $i = 1$ and $j \neq 1$, $X_{1j} = X_{j1} = 0$, and $X_{11} = 0$ by hypothesis. The rank is the number of nonzero diagonal entries, namely $n - 1$. $\square$

Theorem 4 (Family (b): $G = \overline{K_n}$). *Let $n \geq 2$. Then $\omega(\overline{K_n}) = 1$ (`cliqueNum_bot`), so the claimed bound is $n - 2$. The all- (-1) matrix $M = -J$ is admissible for K_n (`isCdV_complete`) and has corank $n - 1$ (`corank_negJ`). Hence $\mu(K_n) \geq n - 1 > n - 2$.*

Proof. (M1) holds since all off-diagonal entries are -1 and all pairs are adjacent. (M3) is immediate: X vanishes on the diagonal and on every edge, hence everywhere. For (M2) and the rank: $M^2 = -nM$, so each eigenvalue λ (with a nonzero eigenvector v) satisfies $(\lambda^2 + n\lambda)v = 0$, i.e. $\lambda \in \{0, -n\}$ (`negJ_eig`). The eigenvalues sum to $\text{tr } M = -n$, so exactly one of them equals $-n$ (`negJ_card`). Thus there is exactly one negative eigenvalue and exactly one nonzero eigenvalue, so $\text{rank } M = 1$ (Mathlib’s `rank_eq_card_non_zero_eigs`). $\square$

Theorem 5 (`conjecture_2149_false`). *(a) For every $n \geq 1$ there is an admissible matrix M for $\overline{K_n}$ with $n - \omega(K_n) - 1 < \text{corank } M$; if $n \geq 2$ then $\text{corank } M = 1$ and $\mu(\overline{K_n}) \geq 1$; and $\mu(\overline{K_n}) \not\leq n - \omega(K_n) - 1$. (b) For every $n \geq 2$ there is an admissible matrix M for $K_n = \overline{\overline{K_n}}$ with $\text{corank } M = n - 1$, $\omega(\overline{K_n}) = 1$, and $\mu(K_n) \not\leq n - \omega(\overline{K_n}) - 1$.*

So the inequality fails for every complete graph with $n \geq 1$ vertices, which is exactly where the conjecture claims tightness, and also for every edgeless graph with $n \geq 2$ vertices, where the bound $n - 2$ is nonnegative and the excess has nothing to do with the sign convention.

4 Lean formalisation and axiom audit

File `lean/Conjecture2149/Basic.lean` (Lean 4 v4.33.1, Mathlib v4.33.1, only `import Mathlib`). Main theorem: `C2149.conjecture_2149_false`. The output of `#print axioms C2149.conjecture_2149_false` is `[propext, Classical.choice, Quot.sound]`. There is no `sorry` and no `native_decide`.

5 Source

Wikipedia, *Colin de Verdière graph invariant*, https://en.wikipedia.org/wiki/Colin_de_Verdi%C3%A8re_graph_invariant: “ $\mu(G)$ is the largest corank of any symmetric matrix $M = (M_{i,j}) \in \mathbb{R}^{(n)}$ such that: (M1) for all i, j with $i \neq j$: $M_{i,j} < 0$ if $\{i, j\} \in E$, and $M_{i,j} = 0$ if

$\{i, j\} \notin E$; (M2) M has exactly one negative eigenvalue, of multiplicity 1; (M3) there is no nonzero matrix $X = (X_{i,j}) \in \mathbb{R}^{(n)}$ such that $MX = 0$ and such that $X_{i,j} = 0$ if either $i = j$ or $M_{i,j} \neq 0$ hold." The same page states " $\mu = -1$ if and only if G is the order-zero graph, K_0 ; $\mu \leq 0$ if and only if G has no more than one vertex". Here $\mathbb{R}^{(n)}$ is read as the space of symmetric real *nimesn* matrices (the page uses it for the symmetric matrix M), so X ranges over symmetric matrices, as in our (M3).